

1141 CS140 資訊概論 Assignment 2

1. Using the instruction set of the simple computer in Section 5.8, write the code for a program that adds n integers typed on the keyboard and displays their sum. You need first to type the value of n . (Hine: use the decrement instructions.)
2. Using the instruction set of the simple computer in Section 5.8, write the code for a program that accepts two integers from the keyboard. If the first integer is 0, the program increments the second integer; if the first integer is 1, the program decrements the second integer. The first integer must be only 0 or 1, otherwise the program fails. The program displays the result of the increment or the decrement.
3. The Fibonacci sequence is the series of numbers where each number is the sum of the two preceding numbers.

$F(0), F(1), F(2), F(3), F(4), F(5), F(6), \dots$

0, 1, 1, 2, 3, 5, 8, ...

Using the instruction set of the simple computer in Section 5.8, write the code for a program that accepts an integer n from the keyboard, the program will calculate the n -th number of the Fibonacci sequence $F(n)$ and display it on the screen.